MAVENS / WEEK 3 PROJECT

Public-source IT & Cybersecurity Research

Project goals, agents, architecture and a practical walkthrough

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4 AGENTS

1 ORCHESTRATOR

REUSABLE MEMORY

GOALS → DELIVERY

Project goals

Turn a company name into a repeatable, source-linked research briefing.

01 Identify comparable companies

Find up to five revenue peers using supported financial evidence and sector checks.

02 Surface recent signals

Review three calendar months of IT, cybersecurity, technology and leadership news.

03 Add professional context

Research CEO, CISO, CTO and CIO candidates, public career history and LinkedIn links.

04 Deliver and remember

Create an editable PowerPoint and retain dated research for comparison and recall.

Architecture: from company input to retained briefing

Explicit workflow routing connects search, local inference, validation and reporting.

COMPANY FORM
Name • website • sector

→

AGENT 1
Revenue peers + checks

→

AGENTS 2 → 3
Search → extract signals

→

AGENT 4
Leadership + background

ORCHESTRATOR → EDITABLE POWERPOINT + RESEARCH JSON

Append-only Data Table memory • Prior-run comparison • Download and recall

Runtime: local n8n + Ollama/Gemma | External evidence: native Google/SerpApi nodes

All agents and their responsibilities

The orchestrator coordinates the four research stages and owns final output.

Agent 1	**Revenue peers**	Company + financial evidence	Up to five eligible peers; evidence gaps
Agent 2	**Public web and news**	Target + selected peers + date window	Dated headlines, links and evidence package
Agent 3	**IT / cyber signal analysis**	Supplied public evidence	Validated, attributed signals and citations
Agent 4	**Leadership background**	Company + saved profile context	Role candidates, career/education claims, links
Orchestrator	**Coordinate, compile, retain**	Peer decisions + signals + profiles	PPTX + research JSON + persistent snapshot

Agent 1: build the revenue peer set

Research Orchestrator canvas; Agent 1 runs in the opening financial stages.



INPUT

Company identity, annual report or retained revenue evidence.

CHECKS

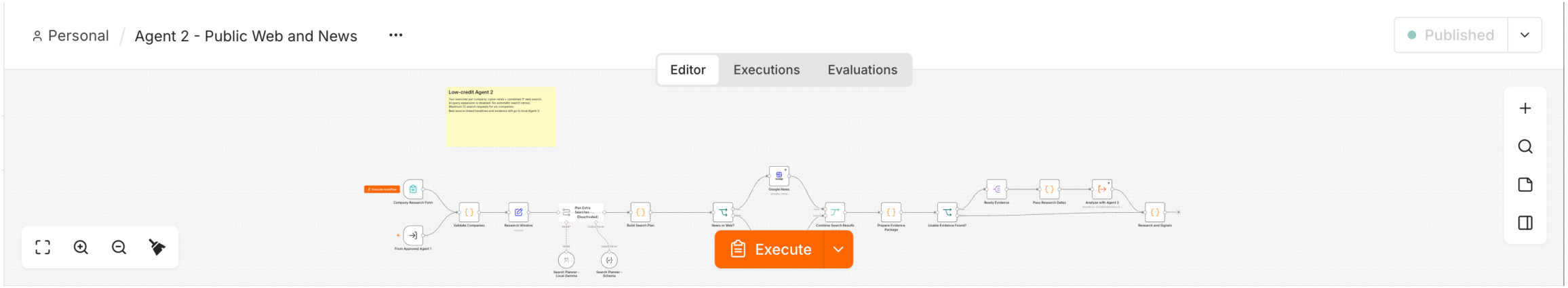
Quoted amount, year, currency, company attribution and range.

OUTPUT

Ranked eligible peers and explicit evidence gaps.

Agent 2: collect public web and news

Native Google News + Google Web return evidence with source URLs.



SEARCH

Current low-credit mode uses two core searches per company.

FILTER

Apply research dates, company matching, URL cleanup and deduplication.

HANDOFF

Package source evidence and send it to Agent 3.

Agent 3: extract supported IT / cyber signals

Local Gemma extracts structured claims; deterministic checks narrow what is retained.



EXTRACT

Leadership, contracts, cyber news and technology signals.

VALIDATE

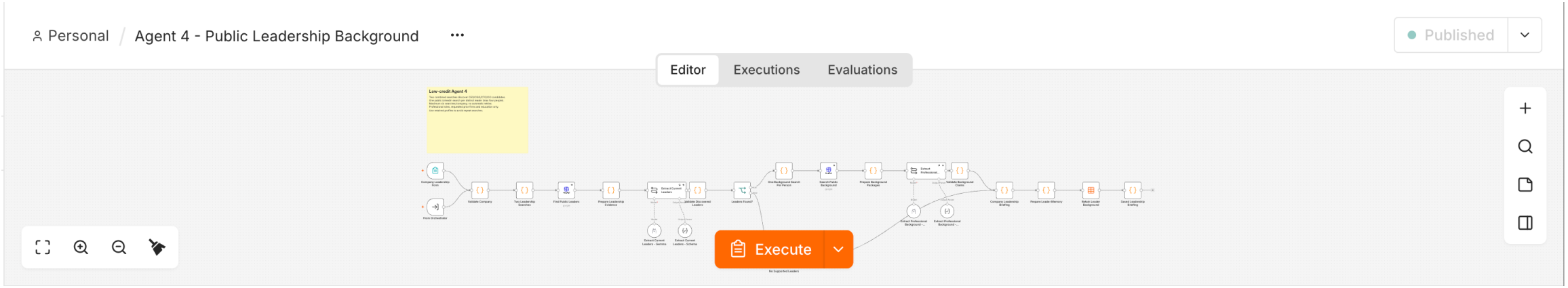
Check company attribution, source quotes and date window.

LABEL

Separate hiring, planned activity and announced deployment.

Agent 4: research public leadership background

Discover CEO / CISO / CTO / CIO candidates and preserve uncertainty.



DISCOVER

Two combined leadership searches, then validate candidates.

ENRICH

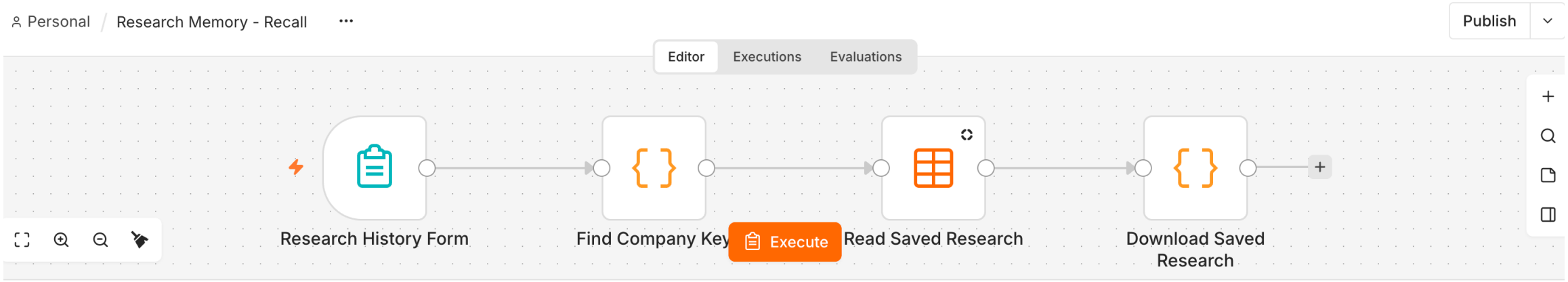
One public LinkedIn search per distinct leader; up to four people.

RETAIN

Append profiles with dates, source links and missing/conflicting roles.

Research memory: retain, compare and recall

Actual recall workflow; production research is retained in the native n8n Data Table.



APPEND

Keep research snapshots, leadership profiles and PowerPoint bytes.

COMPARE

Mark prior evidence and newly observed sources across runs.

RECALL

Search by company; use a run ID to retrieve an older saved version.

How to run: use the existing local installation

Use Research Orchestrator at localhost:5678; the tested fixes are now published.

1	**Start services**	Keep Docker/n8n and local Ollama running; confirm gemma4:12b is available.
2	**Open the workflow**	Open Research Orchestrator, refresh the editor, click Execute workflow and open its form.
3	**Submit the company**	Enter real company name, official website, sector and geography. Attach an annual report when available.
4	**Let the stages complete**	Peer checks → public search → signal extraction → leadership recall/research → presentation and memory.
5	**Download your briefing**	Open Download Briefing → Binary → presentation. Use Research Memory - Recall for saved versions.

Example input: reproduce the Target walkthrough

The main form supplies the company identity and financial-comparison context.

FORM VALUES

Company Name: Target

Official Website: https://corporate.target.com

Industry Sector: Retail

Geography: United States

Annual Report: upload if available

OPTIONAL CONTROLS

Annual Report URL

Minimum / Maximum Revenue Ratio

FX Rates JSON when supported conversion is needed

News window: three calendar months

Expected output: a dated briefing, explicit coverage gaps, source links and a retained research snapshot.

How to set up from GitHub

Import portable workflows, configure local dependencies and reconnect credentials.

01 GET THE PROJECT

Clone amitgandre/Mavens-Week3-AmitG.

Use n8n/saved-workflows/all-workflows.json or import individual production workflows.

02 CONNECT SERVICES

Run n8n with persistent storage.

Install Ollama + gemma4:12b. Configure Ollama and SerpApi credentials; native search nodes must be available.

03 INITIALIZE & TEST

Inspect Research Memory - Setup and create/map the native Data Table.

Publish internal-call workflows, remap references if needed, then run the local model check.

Portable imports are inactive and contain no credential references. Memory and generated files remain local.

Deliverables: a usable research system

Outputs support both presentation and further evidence review.

EDITABLE POWERPOINT

Target-final-briefing.pptx

Peer comparison, selected signals, leadership and linked sources.

COMPLETE RESEARCH

Target-final-briefing.json

Full evidence, quotations, source registry, dates and gaps.

REUSABLE WORKFLOWS

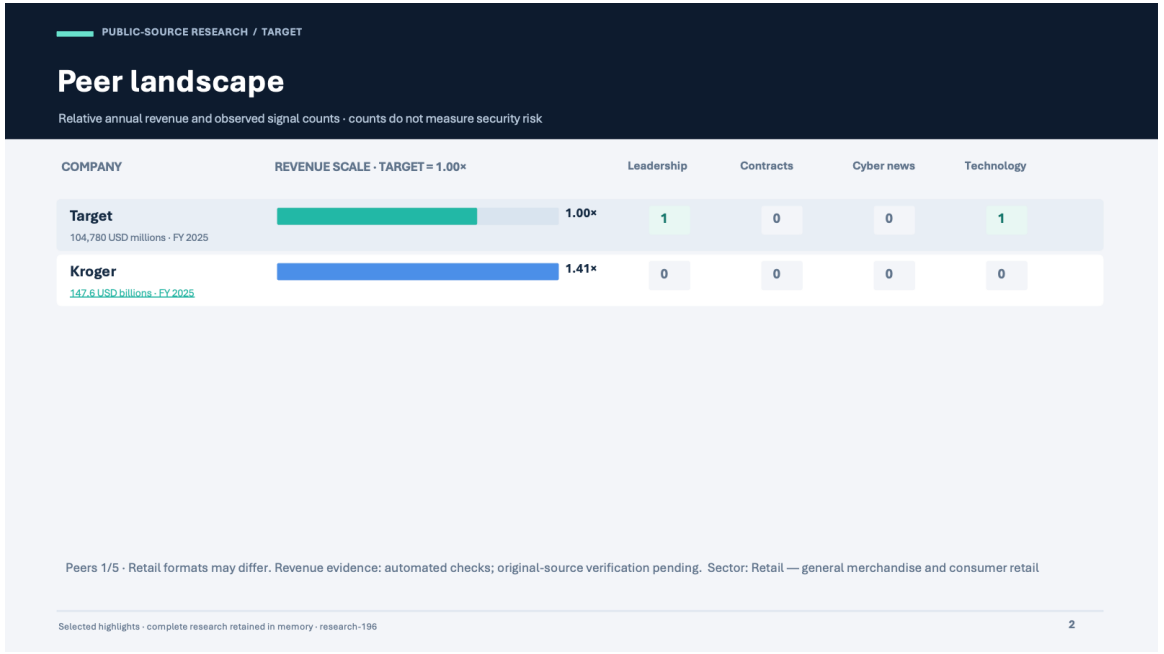
22 portable production/test workflow exports, agent source, checks and documentation.

PERSISTENT HISTORY

Native memory retains dated research and deck bytes; recall retrieves saved versions.

Example delivery: the Target research briefing

Actual generated presentation pages, retained separately from this project walkthrough.



PUBLIC-SOURCE RESEARCH / TARGET

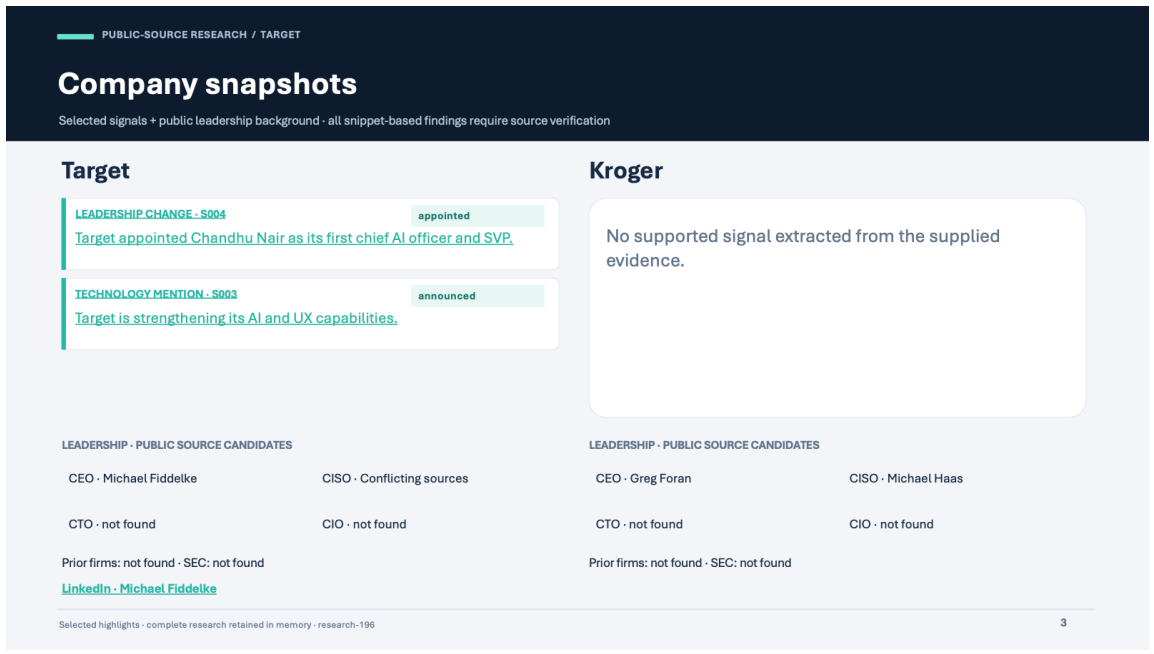
Peer landscape

Relative annual revenue and observed signal counts · counts do not measure security risk

COMPANY	REVENUE SCALE · TARGET = 1.00×		Leadership	Contracts	Cyber news	Technology
Target 104,780 USD millions · FY 2025		1.00×	1	0	0	1
Kroger 147.6 USD billions · FY 2025		1.41×	0	0	0	0

Peers 1/5 · Retail formats may differ. Revenue evidence: automated checks; original-source verification pending. Sector: Retail — general merchandise and consumer retail

Selected highlights · complete research retained in memory · research-196 2



PUBLIC-SOURCE RESEARCH / TARGET

Company snapshots

Selected signals + public leadership background · all snippet-based findings require source verification

Target

LEADERSHIP CHANGE · S004 — appointed

Target appointed Chandhu Nair as its first chief AI officer and SVP.

TECHNOLOGY MENTION · S003 — announced

Target is strengthening its AI and UX capabilities.

LEADERSHIP · PUBLIC SOURCE CANDIDATES

CEO · Michael Fiddelke — CISO · Conflicting sources

CTO · not found — CIO · not found

Prior firms: not found · SEC: not found

LinkedIn · Michael Fiddelke

Kroger

No supported signal extracted from the supplied evidence.

LEADERSHIP · PUBLIC SOURCE CANDIDATES

CEO · Greg Foran — CISO · Michael Haas

CTO · not found — CIO · not found

Prior firms: not found · SEC: not found

Selected highlights · complete research retained in memory · research-196 3

2
Companies

2
Extracted signals

39
Source records

3
Leaders researched

Validation and operating limits

All four agents executed in final run 196; the five-peer coverage target remains unmet.

CHECKED IN THIS PROJECT

Company attribution and quoted evidence

Structured extraction and date handling

Peer eligibility and unsupported-revenue stops

Append-only memory and prior-run recall

PowerPoint opens, exports and fits slide bounds

WHAT THE READER MUST KNOW

Search snippets need original-source review.

Role candidates may be stale or disputed.

Hiring is not evidence of deployment.

Signal counts are not security-risk scores.

Low-credit search limits coverage; missing means insufficient evidence.

Final run 196: all four agents completed; 1/5 peers qualified. Tested changes are published.

Handoff: code, workflow entry points and files

Published workflow, source code and deliverables from the tested implementation.

GITHUB REPOSITORY

github.com/amitgandre/Mavens-Week3-AmitG

Start: README.md

Import: n8n/saved-workflows/README.md

Run / recall: Research Orchestrator; Research Memory - Recall

Presentation source: n8n/project-overview/build_presentation.py

TWO PRESENTATIONS

1 Target research briefing

Business research results

2 Project overview

Goals, agents, architecture, screenshots, operation and deliverables

Open the repository →